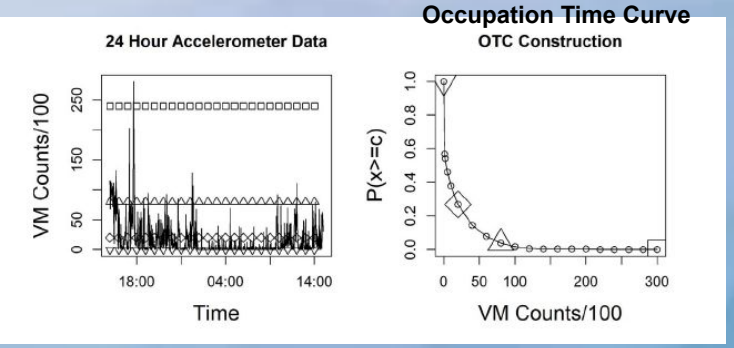
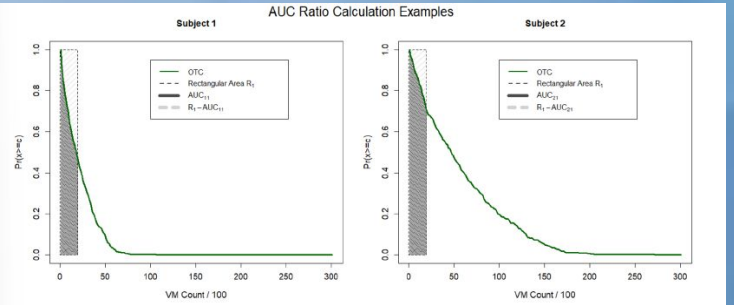
Method:





1.OTC(c)=P(VM(t) ≥ c)= $\frac{\text{duration}\{t:VM(t)\geq c\}}{\text{total duration of } VM(t)}, \qquad c \in [0, C_{\max}]$

Raw data to Functional predictors. OTC summarizes the proportion of time spent above each intensity level, converts noisy time series into smooth, interpretable functional curves.

2.scalar-on-function regression:

$$A_{ij} = \int_{c_{j-1}}^{c_j} OTC_i(c)\, dc,$$

$$y_i \sim \sum_{j=1}^{J} \theta_j A_{ij} + Z_i^\top \alpha$$

Discretize intensity domain into J fine bins. Approximate the integral using AUC features. Leads to a high-dimensional linear regression in Aij.

3.Window Detection via L0 Regularization & MIO:

Assume the coefficient function θ(c) is piecewise constant

Activity effects are constant within contiguous intensity windows

Impose an L0 penalty on coefficient changes to control window number

Solve using mixed integer optimization (MIO)

Outputs data-driven activity windows and their health effects

Data:

```
Host: tri-login01.scinet.local

SciNet welcomes you to the Trillium cluster
at the University of Toronto!

Docs: https://docs.alliancecan.ca/wiki/Trillium
Help: trillium@tech.alliancecan.ca

This is the login node tri-login01. Use this node to develop and compile code,
to run short tests, and to submit CPU compute jobs to the scheduler.

For help with your transition to Trillium, please see:
https://docs.alliancecan.ca/wiki/Transition_from_Niagara_to_Trillium

Prefer a self-guided course instead?
Check out: see https://scinet.courses/1389

Welcome xiankai, your access to this system has been logged.
If you are not xiankai, please disconnect immediately.

[xiankai@tri-login01 ~]$ cd $HOME

ln -s /project/rrg-oespinga/oespinga/OAI/ClinicalData OAI_data

[xiankai@tri-login01 ~]$ ls ~

links  OAI_data  R
[xiankai@tri-login01 ~]$
```

```
{r load datasets}
# wd <- "~/Downloads/OAI_project/ClinicalData/OAICompleteData_ASCII"

wd <- "/project/rrg-oespinga/oespinga/OAI/ClinicalData/CompleteData_ASCII/"

accel6=read.table(paste0(wd,"/Accelerometry06.txt"), sep="|", header=T)
accel8=read.table(paste0(wd,"/Accelerometry08.txt"), sep="|", header=T)

day6=read.table(paste0(wd,"/AccelDataByDay06.txt"), sep="|", header=T)
day8=read.table(paste0(wd,"/Acceldatabyday08.txt"), sep="|", header=T)

min8=read.table(paste0(wd,"/Acceldatabymin08.txt"), sep="|", header=T)
min6=read.table(paste0(wd,"/Acceldatabymin06.txt"), sep="|", header=T)
```

Acceldatabymin06

Acceldatabymin08

same ID: same individual

```
> levels(min8$VERSION)
[1] "8.2.1"
> levels(min6$VERSION)
[1] "6.2.2"
```

time index (t)

The core variable: VM(t), which is the activity count

```
> summary(min8)
      ID            VERSION          V08PAStudyDay  V08MinSequence
 Min.   :9000099   Length:14878346   Min.   : 781   Min.   :   1.0
 1st Qu.:9258864   Class :character  1st Qu.: 948   1st Qu.: 360.0
 Median :9503396   Mode  :character  Median :1034   Median : 720.0
 Mean   :9503245                     Mean   :1040   Mean   : 720.3
 3rd Qu.:9742313                     3rd Qu.:1146   3rd Qu.:1080.0
 Max.   :9999878                     Max.   :1427   Max.   :1440.0

 V08SuspectMinute     V08MINCnt        V08PAWeekDay          V08PAMonth
 Min.   :0.000000   Min.   :    0.0   Length:14878346    Min.   : 1.00
 1st Qu.:0.000000   1st Qu.:    0.0   Class :character   1st Qu.: 4.00
 Median :0.000000   Median :    0.0   Mode  :character   Median : 7.00
 Mean   :0.001914   Mean   :  137.6                      Mean   : 6.92
 3rd Qu.:0.000000   3rd Qu.:   29.0                      3rd Qu.:10.00
 Max.   :1.000000   Max.   :20758.0                      Max.   :12.00
                    NA's   :73
```

```
> summary(min6)
      ID            VERSION          V06PAStudyDay  V06MinSequence
 Min.   :9000099   6.2.2:20629910   Min.   : 28.0   Min.   :   1.0
 1st Qu.:9266787                    1st Qu.:202.0   1st Qu.: 360.0
 Median :9513843                    Median :295.0   Median : 720.0
 Mean   :9506945                    Mean   :304.6   Mean   : 720.1
 3rd Qu.:9742871                    3rd Qu.:413.0   3rd Qu.:1080.0
 Max.   :9999878                    Max.   :707.0   Max.   :1440.0

 V06SuspectMinute     V06MINCnt        V06PAWeekDay          V06PAMonth
 Min.   :0.0000000   Min.   :    0.0   Length:20629910    Min.   : 1.000
 1st Qu.:0.0000000   1st Qu.:    0.0   Class :character   1st Qu.: 3.000
 Median :0.0000000   Median :    0.0   Mode  :character   Median : 6.000
 Mean   :0.0006849   Mean   :  143.1                      Mean   : 6.489
 3rd Qu.:0.0000000   3rd Qu.:   34.0                      3rd Qu.:10.000
 Max.   :1.0000000   Max.   :20670.0                      Max.   :12.000
                     NA's   :18
```

#	Variable	Type	Len	Format	Informat	Label
1	ID	Char	7	$7.	$7.	ReleaseID
2	VERSION	Char	5			Release Version
3	V08PAStudyDay	Num	8			PA study day (range 781-1427, with Day 1=August 01,2008)
4	V08MinSequence	Num	8			Sequential daily obs number (range 1-1440, represents 00:00 to 23:59 minutes)
5	V08SuspectMinute	Num	8			Suspicious minute suggesting monitor wear during sleep
6	V08MINCnt	Num	8			Activity counts
7	V08PAWeekDay	Char	9	$9.		Day of week
8	V08PAMonth	Num	8			PA monitoring month

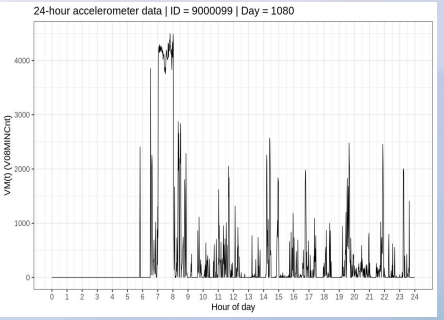
```
$ ID          <int> 9000099, 9000099, 9000099, 9000099, 9000099, 9000099,…
$ VERSION     <chr> "8.2.1", "8.2.1", "8.2.1", "8.2.1", "8.2.1", "8.2.1",…
$ V08PAStudyDay <int> 1080, 1080, 1080, 1080, 1080, 1080, 1080, 1080, 1080,…
$ V08MinSequence <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16…
$ V08SuspectMinute <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,…
$ V08MINCnt   <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,…
$ V08PAWeekDay <chr> "Saturday", "Saturday", "Saturday", "Saturday", "Satu…
$ V08PAMonth  <int> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7,…
```

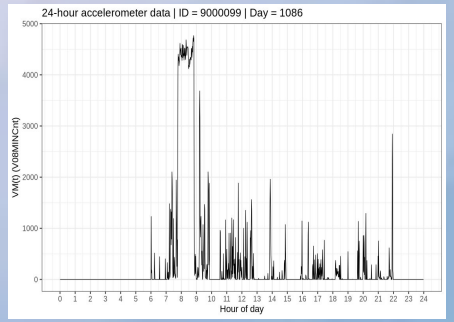
might need to be put in measured towards the duration of minutes

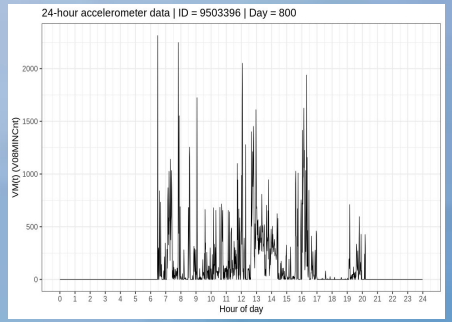
could be used to further investgation, such as: weekly or seasonally

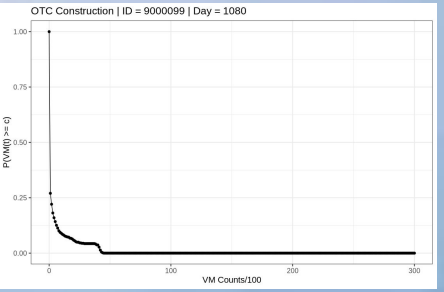
```
> levels(min6$V06PAWeekDay)
[1] "Friday"    "Monday"    "Saturday"  "Sunday"    "Thursday"  "Tuesday"
[7] "Wednesday"
```

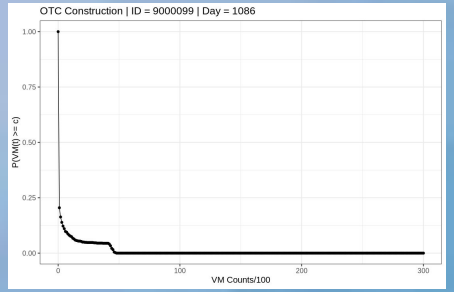
Current Step:

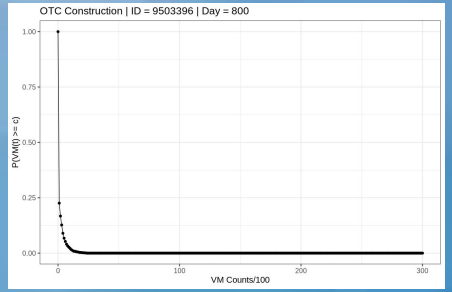












Future Step: